

The background of the slide is a dark blue field filled with a complex network of glowing blue lines and dots, resembling a digital or social network. The dots vary in size and brightness, and the lines connect them in a web-like pattern, creating a sense of depth and connectivity.

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Influencer-Fraud Detection

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AGENDA:

The agenda of the fraud-influencer detection project is scoring **INSTAGRAM** and **TIKTOK** handles for fake engagement using public metrics and unsupervised machine learning.



INTRODUCTION & PROBLEM STATEMENT:

- . Define influencer frauds (fake followers, bought likes, abnormal growth)
- . Explaining why detecting fake engagement matters for brand, advertisers, and platforms
- . Platforms like **INSTAGRAM** and **TIKTOK** host millions of influencers who collaborate with companies to promote products and services. Brands invest heavily in influencer partnerships, often spending thousands of dollars per campaign, expecting genuine reach and authentic engagement.

TITLE:

- . Anomaly Detection of Fraudulent Influencers on Social Media
- . Machine Learning-Based Detection of Fake Engagement in Influencer Profiles
- . Unsupervised Learning Approach to Identify Influencer Fraud
- . Fraud Risk Scoring System for **INSTAGRAM** and **TIKTOK** Influencers
- . Automated Detection of Influencer Engagement Fraud Using Data-Driven Models
- . The real Vs The fake: AI to Unmask Fraudulent Influencers.

OBJECTIVES:

- . Build an anomaly-detection pipeline to flag likely fraudulent influencer accounts
- . Produce a calibrated 0-100 fraud risk score per handle
- . Meet business requirement: $\geq 85\%$ precision on 'FRAUD' in the 200 labeled calibration set
- . Deliver: Jupyter notebook, influencer_scores.csv, and a pdf report with thresholds & business rules.

How to Detect Fake Followers and Engagement?



DATA & SCHEMA:

Collect 5,000 influencer profiles (socialBlade API or simulated). Each row should include:

- . handle (string)
- . platform (Instagram/ TikTok)
- . followers (int)
- . avg_likes (int)-mean likes per recent post
- . avg_comments (int)
- . posts_30d (int)
- . follower_growth_30d (int)
- . sample_post_dates (optional)



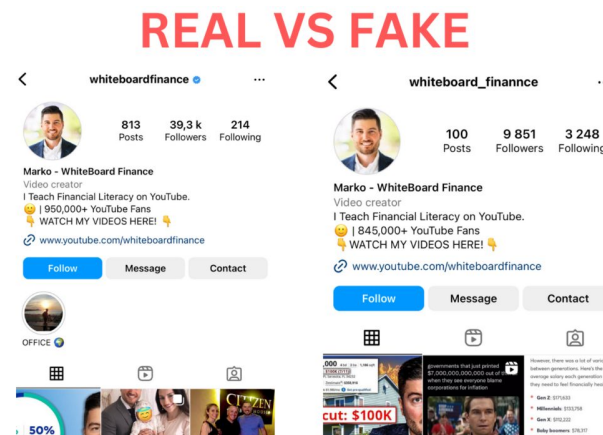
Save labeled sample: labeled_200.csv with handle+label (0=genuine,1=fraud).

ENGINEER FEATURES:

- . $\text{engagement_per_follower} = (\text{avg_likes} + 2 * \text{avg_comments}) / \text{followers}$
- . $\text{likes_per_post} = \text{avg_likes} / \max(1, \text{posts_30d})$
- . $\text{comments_per_posts} = \text{avg_comments} / \max(1, \text{posts_30d})$
- . $\text{growth_rate_30d} + \text{followers_growth_30d} / \max(1, \text{followers})$
- . $\text{likes_followers_ratio_log} = \log_{1p}(\text{avg_likes}) - \log_{1p}(\text{followers})$
- . optional time-series features: week-over-week growth, posting cadence variance.

MODELING APPROACH:

- . **Primary models:** Isolation Forest (IF)+ Local Outlier Factor (LOF).
- . Preprocessing: StandardScaler on numeric features.
- . **Ensemble:** Convert model outputs to anomaly probabilities (rank-based), then weighted average (example: 0.6 IF + 0.4 LOF).
- . **Score mapping:** anomaly_prob → fraud_score_0_100(round to two decimals).



COLLABORATION & THRESHOLDING:

- . Use the 200 labeled profiles only to choose a decision threshold on `fraud_score_0_100`.
- . **Procedure:** for candidate thresholds (e.g., 50-99), compute precision and recall on `labeled_200`; choose the smallest threshold that archives precision ≥ 0.85 while maximizing recall; if none do, choose threshold with highest precision and document tradeoffs.
- . Always report chosen threshold, precision, recall, predicted positives count.

FAKE INFLUENCER DATASETS

1. Public Datasets

- . Fake Followers Detection Dataset (**INSTAGRAM/TIKTOK**)- Kaggle has versions where labels exists (genuine vs fake).
- . **INSTAGRAM** fake engagement datasets (some small open-source research datasets exist).

2. Synthetic Data

- . Generate profiles with unrealistics ratios (e.g., 1million followers but <0.01% engagement, sudden 50% growth in one day, 90% follow-back ratio).
- . Label them as “fraud” for testing,

3. Internal Labeled Sample

- . Your 200 manually-labeled profiles can act as a mini validation set.
- . Split into collaboration (train threshold) vs holdout (validation).

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How to Spot Fake Followers

XOMAD



Content Quality

If an influencer has posted multiple giveaways, chances are they have fake followers. Make sure the quality of their content matches their follow count.



Growth Rate

Check Social Blade. Large spikes in followers or a constant growth of 5,000 a week are fishy. Growth should be constant with small spikes throughout.



Follower Count vs. Engagement Rate

The industry standard for Instagram engagement is about 2% average. Anything lower than that is a possible red flag.



Comments from Other Influencers

Comments left by other influencers could be genuine, or they could be from an Instagram pod, which are created to increase engagement.



Engagement Per Type of Post

A selfie should receive higher engagement versus a sponsored post. Consistent engagement on every type of post is something to be cautious of.



Location of Followers

Having too many followers from foreign countries, like India, or countries where the influencer has never visited can be an indicator of bought followers.

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APPLY YOUR PIPELINE:

1. Run the **feature engineering** script to transform raw fake datasets into features.
2. Apply your **Isolation Forest+LOF ensemble** to compute anomaly scores.
3. Convert anomaly scores → `fraud_score_0_100`.

COMPARE SCORES VS LABELS:

On the validation datasets

(known_fake_set.csv with labels):

- . Compute **precision, recall, F1** on the fraud class.
- . Report metrics at the chosen threshold ($\geq 85\%$ precision requirement).

Example metrics to compute:

```
from sklearn.metrics import classification_report, confusion_matrix, precision_score, recall_score
```

```
print("Precision:", precision_score(y_true, y_pred))
```

```
print ("Recall:", recall_score(y_true, y_pred))
```

```
print ("F1:", f1_score(y_true, y_pred))
```

BUSINESS INTERPRETATION:

.If precision $\geq 0.85\%$ meets requirement

. If recall is low (e.g. 40-50%), highlight that the tool is **conservative** (catches fewer fakes but with high confidence).

. Document **false positive cases** - sometimes creators look anomalous but are genuine (e.g., viral micro-influencers).

CONTINUOUS VALIDATION:

- . Keep a rolling datasets of vetted influencers (manual reviewers confirms fraud/genuine).
- . Re-validate monthly to ensure performance doesn't degrade as Fraud tactics evolve.

CONCLUSION:

- . Built an anomaly-detection tool for **INSTAGRAM&TIKTOK** influencers.
- . Generated a 0-100 fraud risk score using engagement and growth features.
- . Applied isolation forest & local outlier factor, calibrated on a labeled sample.
- . Achieved $\geq 85\%$ precision on detecting fraudulent accounts.
- . Delivered actionable vetting recommendations for flagged influencers.



The background is a dark blue gradient with a complex network of glowing blue lines and dots, resembling a molecular structure or a data network. The lines and dots are more prominent in the lower half of the image, creating a sense of depth and connectivity.

Thank You!